

Haozhe Luo

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[\[Google Scholar\]](#)

EDUCATION

Northeastern University, Qinhuangdao, China

2023 – 2027

- *B.Eng. Candidate* in Communication Engineering, expected June 2027
- **GPA: 3.61 / 5**

RESEARCH INTERESTS

- LLM Reasoning
- AI Agents
- Multimodal Intelligence

RESEARCH EXPERIENCE

Mechanist: Autonomous Research Agent for LLM Interpretability [\[HF Paper\]](#) [\[Website\]](#) May 2026 – Present

- Co-Developed Mechanist, the first AI Scientist specifically designed for LLM mechanistic interpretability, to discover novel behavioral phenomena in LLMs and explain their underlying neural/cognitive mechanisms.
- Engineered a massive knowledge base (43M+ papers, 157M entities, 3B triplets) and a 6-stage automated pipeline with 20 mechanism-probing tools and Human-in-the-Loop workflows, enabling closed-loop scientific discovery.
- Leveraged Mechanist to discover and analyze the Subliminal Learning phenomenon in LLMs, uncovering implicit information transfer patterns and behavioral adaptation through systematic mechanistic interventions.

MemTrapBench: Benchmarking Cognitive Traps in LLM Long-Term Memory Use Jan 2026 – Present

Co-First Author | Under Review at AAAI 2027 [\[HF Paper\]](#) [\[Github\]](#)

- Formulated MemTrapBench: Benchmarked cognitive traps in LLM long-term memory, showing even relevant memories cause >10% performance drops via reasoning fixation and belief distortion.
- Dataset & Framework: Built a multi-turn generation pipeline with 2-stage quality control, creating 1,050 test cases across 4 scenarios (Task Boundary, Cognitive Bias, Trauma, Safety).
- Proposed AdaptiveMem: Developed an inference-time prompt skill that detects and bypasses memory traps, boosting accuracy by up to 14.9% across memory frameworks without degrading standard benchmark performance.
- The research has received considerable attention on X, reaching 30k+ views, 300+ likes, and 50+ reposts.

SocialReasonBench: Benchmarking Higher-Order Social Reasoning in LLMs Nov 2025 – May 2026

Co-First Author | Accepted at EMNLP 2026 [\[OpenReview\]](#)

- Proposed SocialReasonBench, a novel video-QA benchmark comprising 532 complex instances to evaluate higher-order social reasoning, causal inference, and counterfactual reasoning in large multimodal models.
- Developed a multi-agent data curation pipeline (Director, Tracker, and Generator agents) that automated clip localization, game-state grounding across 4 grounding tiers, achieving 100% entity anonymization to prevent benchmark contamination.
- Introduced a theory-driven evaluation framework spanning 4 social interaction types, 7 reasoning dimensions, and 5 diagnostic trap mechanisms; successfully benchmarked 8 leading LLMs, uncovering critical reasoning blind spots (e.g., GPT-4o's accuracy drops from 85.14% in intent recognition to 50.00% in counterfactual reasoning).

Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies Sep 2025 – Nov 2025

Accepted at EMNLP 2026 [\[HF Paper\]](#)

- Developed WereBench and WereAlign framework, processing 100+ hours of video into a 32.4M-token multimodal dataset to benchmark LLMs against winning MVPs' strategies across 5 social dimensions.
- Evaluated dozens of frontier LLMs (e.g., Gemini-2.5-Pro, DeepSeek-R1), executing RR/OSR interventions to prove that LLMs are highly susceptible to human rhetorical manipulation in strategic environments.

PROJECTS

Hardware-Software Co-Designed AI Vision-Based Online Inspection System for Under-Rail Components

- End-to-End System Integration: Partnered with China Railway Shanhaiguan Bridge Group to design and deploy a dual-end AI vision system for Rail Base Plates, combining factory 3D measurement and low-power (99%+ self-sufficient) in-track monitoring.
- Sub-Pixel 3D Metrology: Formulated an Attention-UNet-based structured-light reconstruction engine, yielding 0.01 mm dimensional accuracy, 0.2-pixel center extraction, and ± 0.08 mm repeatable precision.

- **Environmental Adaptation:** Programmed a multi-spectral sensing controller running DDPG reinforcement learning for adaptive illumination, improving environmental adaptability by 62%, noise resistance by 28%, and ensuring <600 ms real-time latency.
- **High-Impact Outcomes & Commercialization:** Authored 2 patent applications (in substantive exam) and 5 software copyrights; won a National First Prize in the AI Algorithm Elite Competition and a National Bronze Medal in the China International College Students' Innovation Competition. Successfully secured a RMB 5M investment intent agreement from a leading technology enterprise, following a prior RMB 600K commercial contract.

Deep Reinforcement Learning and Vision-Based Gomoku Robotic System

- Designed and implemented the game engine, evolving from Alpha – Beta pruning to an AlphaZero-style self-play reinforcement learning framework with distributed search, winning the Second Prize in the China Undergraduate Computer Game Tournament.
- Integrated a YOLOv5-based vision module achieving 98.2% piece recognition accuracy, supporting reliable real-time board-state reconstruction and closed-loop robotic operation.
- Built an end-to-end vision-controlled Gomoku robot implementing closed-loop state perception and stone placement; exhibited at the 2024 Beijing-Tianjin-Hebei AI Expo.

HONORS & AWARDS

International & National

- 2025: Bronze Award, China International College Students' Innovation Competition
- 2025: First Prize, Global Campus AI Algorithm Elite Competition
- 2025: Second Prize, National Computer Game Playing Tournament
- 2025: Second Prize, TRIZ Cup Innovation Method Competition
- 2026: Second Prize, National Energy-Saving and Emission Reduction Competition
- 2024: Third Prize, APMCM Asia-Pacific Mathematical Contest in Modeling

Provincial

- 2026: First Prize, National Energy-Saving and Emission Reduction Competition
- 2025: Gold Award, China International College Students' Innovation Competition
- 2024: Bronze Award, China International College Students' Innovation Competition

University-Level

- 2024,2025: Innovation and Entrepreneurship Scholarship, Northeastern University
- 2025: Outstanding Student , Northeastern University
- 2025: Outstanding Individual, Summer Social Practice Program , Northeastern University
- 2025: 3rd Place in High Jump, 39th University Sports Meeting, Northeastern University
- 2025: Sports Star, Northeastern University

INTERNSHIP

ZJUNLP LAB – Zhejiang University

Jan 2026 – Present

AI Research Intern Supervisor: Prof. Ningyu Zhang

UTS NLP – University of Technology Sydney

Remote Oct 2025 – Mar 2026

AI Research Intern Supervisor: Prof. Ling Chen

HALO LAB – Mohamed bin Zayed University of Artificial Intelligence

Remote Sep 2025 – Nov 2025

AI Research Intern Supervisor: Prof. Xiuying Chen

TECHNICAL SKILLS

- Languages: Mandarin (Native), English (Proficient – CET6 549)
- Programming Languages: Python, C++, CUDA , Matlab
- Machine Learning & AI: PyTorch, Transformers, LLMs, LMMs, Reinforcement Learning, AI Agents
- Tools & Frameworks: Git/GitHub, Linux, Hugging Face Transformers, Next.js, Zotero